



Loops

Multiple edges

~~a_{22}~~ $a_{22} = 1$

$$a_{12} = 2 \quad \left([1, 2], [1, 2], [2, 2] \right)$$

Simple graph - no mult edges

- no loops

- Undirected (no arrows)

Adjacency Matrix

only 0, 1

Diagonal $\rightarrow a_{ii} = 0$

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Undirected

$\Leftrightarrow$

matrix

is Symmetric



$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

Each edge connects 2 vertices



if G is simple then $x \neq y$

if G is simple at most one edge
from x to y

there is a function F

$F: \text{Edges} \rightarrow \text{Pairs of Vertices}$

By def of edge, e corresponds
to $\{x, y\}$ but also to $\{y, x\}$
as ordered lists

$[x, y]$ or $[y, x]$

$F: \text{Edges} \rightarrow \text{Subsets of } V$
of size 2

Well-defined because x, y or y, x

and $x \neq y$ no repeats Same set
Unordered list w/ no repeats is a set

1. The first part of the paper is devoted to a general discussion of the problem.

2. In the second part, we shall consider the case of a single particle.

3. The third part is devoted to the case of a system of particles.

4. In the fourth part, we shall discuss the results of our calculations.

5. The fifth part is devoted to a summary of the results.

6. The sixth part is devoted to a discussion of the physical meaning of the results.

7. The seventh part is devoted to a discussion of the experimental results.

8. The eighth part is devoted to a discussion of the theoretical results.

9. The ninth part is devoted to a discussion of the results of our calculations.

10. The tenth part is devoted to a summary of the results.

11. The eleventh part is devoted to a discussion of the physical meaning of the results.

12. The twelfth part is devoted to a discussion of the experimental results.

13. The thirteenth part is devoted to a discussion of the theoretical results.

14. The fourteenth part is devoted to a discussion of the results of our calculations.

15. The fifteenth part is devoted to a discussion of the physical meaning of the results.

16. The sixteenth part is devoted to a discussion of the experimental results.

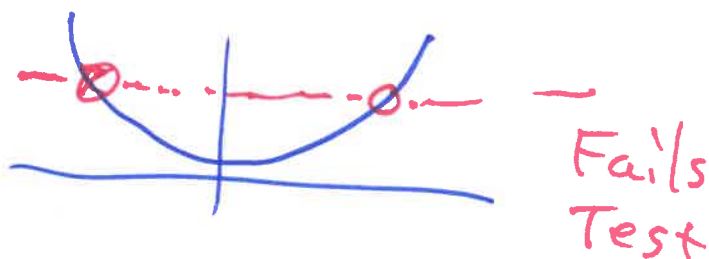
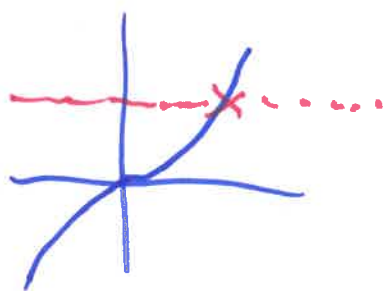
$$F: A \rightarrow B$$

is 1-1 or injective

$$\text{if } F(a_1) = F(a_2)$$

$$\text{then } a_1 = a_2$$

Horizontal line test



$\forall b \in B$, either $\exists ! a$ s.t. $F(a) = b$
or no a s.t. $F(a) = b$

If G is simple then

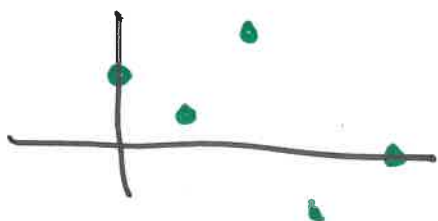
is

$F: \text{Edges} \rightarrow \text{Subset of } V \text{ of size } 2$

injective?

Given $\{x, y\}$

either 1 edge
or 0 edges



Injective 1-1

Q: $\exists$ G w/ 7 vertices

each w/ degree 3 (Regular Graph)

Each edge has 2 vertices

Contribution of one edge
to the $\sum_{i=1}^7 \deg(V_i)$ is +2

Introduce notation.

Let $V_i = i$ th vertex
 i ranges 1 to 7

Call the total degree = $\sum_{i=1}^7 \deg V_i$

$E = \#$ edges

$\alpha =$

$\alpha = 2E \leftarrow \text{even number}$

1900

1901

1902

1903

1904

1905

1906

1907

1908

1909

1910

$\deg v_i = 3$ given in problem

$$\text{So } \alpha = \sum_{i=1}^7 \deg v_i = \sum_{i=1}^7 3 = 7 \cdot 3 = 21$$

Therefore 21 is even

Contradiction.

Go back to beginning

Make a claim

There is no simple graph
with 7 vertices and every vertex
has $\deg 3$.

Start a Proof by Contradiction

Proof by Contradiction.

Assume G is a simple graph
w/ 7 vertices v_i $i=1, \dots, 7$
and $\deg v_i = 3 \forall i$

1. The first part of the paper is devoted to a discussion of the general principles of the theory of the structure of the atom.

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$$A(2,2) = A(m-1, A(m, n-1))$$

when $m=2$ and $n=2$

$$= A(1, \underline{A(2, 1)})$$

$$\underline{A(2, 1)} = \underline{2}$$

$$A(2, 2) = A(1, 2)$$

$$A(1, 2) = A(m-1, A(m, n-1)) \\ = A(0, A(1, 1))$$

$$A(1, 1) = 2$$

$$A(1, 2) = A(0, 2)$$

$$A(0, 2) = 2n = 2 \cdot 2 = 4$$

$$A(1, 2) = 4$$

$$A(2, 2) = 4$$

